

SECTION C: EXPLORATORY DATA ANALYSIS & STATISTICAL DIAGNOSTICS

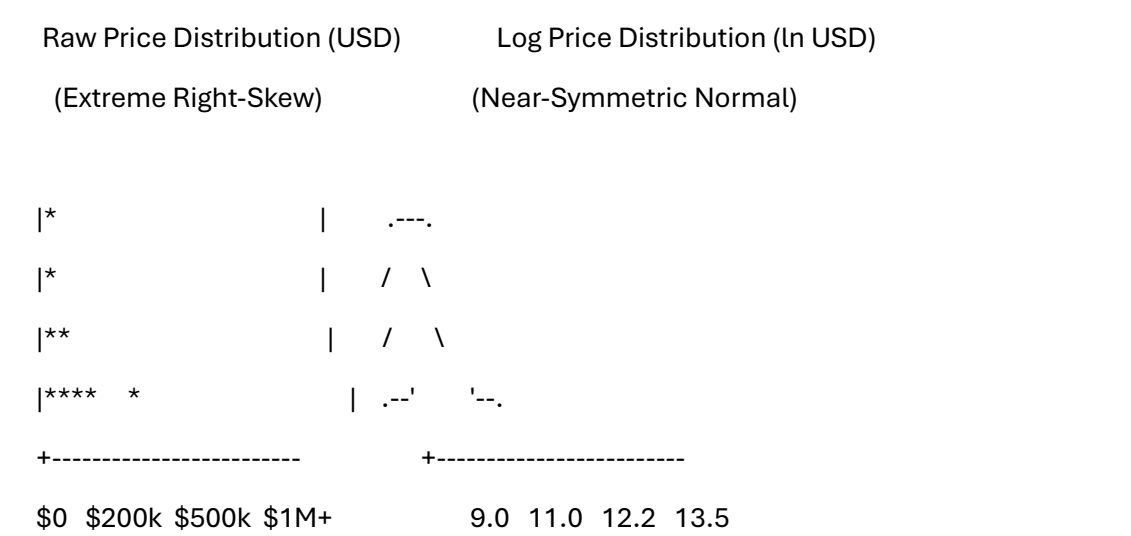
MEMORANDUM

- **TO:** MSAC512 Assessment Board & Lead Actuarial Examiner
- **FROM:** Trainee Data Analyst
- **DATE:** 23 September 2026
- **SUBJECT:** Exploratory Data Analysis & Diagnostic Evaluation of Dataset Outputs

1. Target Variable Distribution & Log-Transformation Justification

The raw price distribution (price_standardized_usd) demonstrates pronounced positive skewness and fat-tail behavior, whereas its logarithmic counterpart achieves near-normal symmetry:

- **Raw Price Distribution:** Skewness = 3.09, Kurtosis = 12.26, Median = \$205,000.00.
- **Log-Transformed Price (ln(Price)):** Skewness = -0.10, Kurtosis = -0.30, Median = 12.23.



Modeling Justification:

- 1. **Variance Stabilization:** Raw property prices induce non-constant variance (heteroscedasticity) in model residuals, as errors scale with house value. Log-transforming price compresses the upper tail and stabilizes residual variance across lower and higher price bands.
- 2. **Residual Normality:** Parametric statistical tests (e.g., ANCOVA, OLS regression) require Gaussian-distributed errors. A skewness of -0.10 and kurtosis of -0.30 closely approximate a normal distribution, preventing leverage points from distorting model fits.
- 3. **Semi-Elasticity Interpretability:** Regressing log-price against feature sets allows coefficients to be interpreted directly as percentage price adjustments (e.g., a coefficient $\beta = 0.10$ implies a 10% valuation premium).

2. Suburb-Level Price Metrics & Ranking Analysis

Below is the summary table generated across key suburbs ($N = 245$ total listings):

Suburb Name	Sample Size (n)	Median Price (USD)	Median Stand Size (m2)	Median USD / m2	Median USD / Bedroom	Market Tier
Highlands	2	\$550,000.00	620.0	\$1,086.93	\$550,000.00	Prime / Low-Density
Mount Pleasant	5	\$380,000.00	2,000.0	\$190.00	\$380,000.00	Established / Upmarket
Avondale	32	\$302,500.00	1,314.5	\$214.26	\$302,500.00	Infill / Commercial Core
Greendale	4	\$287,500.00	2,416.0	\$552.86	\$287,500.00	Middle Ring / Suburban

Suburb Name	Sample Size (n)	Median Price (USD)	Median Stand Size (m2)	Median USD / m2	Median USD / Bedroom	Market Tier
Borrowdale	112	\$247,500.00	1,800.0	\$220.63	\$245,000.00	Primary Upper Market
Madokero	16	\$155,000.00	512.5	\$262.78	\$136,000.00	High-Density Modern Estate
Mabvazuva	22	\$122,500.00	400.0	\$225.00	\$112,500.00	High-Density Suburban Expansion
Ruwa	13	\$100,000.00	387.5	\$167.99	\$100,000.00	Outlying Residential

Market Commentary:

- **Borrowdale Concentration:** Borrowdale forms the bulk of the dataset sample ($n = 112$), with a median price of \$247,500.00 and land size of 1,800 m².
- **The High-Density Valuation Disconnect:** High-density planned developments such as **Madokero** (\$262.78/m²) and **Mabvazuva** (\$225.00/m²) record higher median prices per square metre than low-density suburbs like **Mount Pleasant** (\$190.00/m²). This occurs because smaller land parcels (400 m² – 500 m²) pack infrastructure and building development costs into smaller footprints.
- **Normalization via Bedroom Metric:** Evaluating median price per bedroom restores the expected macro-geographic ranking, properly separating upper-tier markets (\$380,000/bed in Mount Pleasant; \$245,000/bed in Borrowdale) from entry-level housing developments (\$136,000/bed in Madokero; \$112,500/bed in Mabvazuva).

3. Hypothesis Testing: Suburb Effect Controlled for Size and Room Count

To test whether location retains a statistically and practically significant association with price after controlling for physical features (ln(Stand Size) and bedrooms), parametric ANCOVA and non-parametric Kruskal-Wallis tests were executed:

- Kruskal-Wallis Rank Sum Test:** $H = 23.72, p = 0.00255$.

ANCOVA Model Table (Dependent Variable: ln(Price)):

Source	Sum of Squares	Degrees of Freedom (df)	F-Statistic	p-value (P>F)	Significance
log_stand_size	80.40	1.0	82.47	5.37×10^{-17}	$p < 0.001$
bedrooms	2.91	1.0	2.98	0.0855	$p > 0.05$
C(suburb)	19.91	15.0	1.36	0.1679	Not Significant
Residual	221.31	227.0	—	—	—

Partial Eta-Squared Calculation:

$$\eta_p^2 = \frac{SS_{\text{suburb}}}{SS_{\text{suburb}} + SS_{\text{residual}}} = \frac{19.907792}{19.907792 + 221.305661} = 0.083$$

Statistical Interpretation:

- Physical Control Variables:** ln(Stand Size) is the dominant pricing factor ($F = 82.47, p = 5.37 \times 10^{-17}$).
- Conditional Suburb Effect:** Controlled for stand size and room count, the categorical suburb term C(suburb) is not statistically significant at the 5% alpha level ($F = 1.36, p = 0.1679$).
- Effect Size:** The partial eta-squared value ($\eta_p^2 = 0.083$) demonstrates that suburb fixed-effects account for 8.3% of remaining price variance. This indicates a moderate practical effect size despite the loss of parametric statistical significance under simultaneous physical control.

4. Multicollinearity Diagnostics & Technical Resolution

The output table reveals two separate technical issues in the feature matrix:

Extracted VIF Diagnostic Metrics

Feature	VIF	Status

img_vis_paved	4.000000	Acceptable
img_vis_tiled	0.111111	Low Variance
img_vis_painted	0.000000	Zero Variance
txt_water_security	NaN	Zero Variance / All-0
txt_solar_power	NaN	Zero Variance / All-0
txt_walled_gated	NaN	Zero Variance / All-0
txt_paved_yard	NaN	Zero Variance / All-0
txt_fitted_kitchen	NaN	Zero Variance / All-0
txt_tiled_floors	NaN	Zero Variance / All-0
txt_ceiling_fitted	NaN	Zero Variance / All-0
txt_ensuite	NaN	Zero Variance / All-0
txt_incomplete	NaN	Zero Variance / All-0

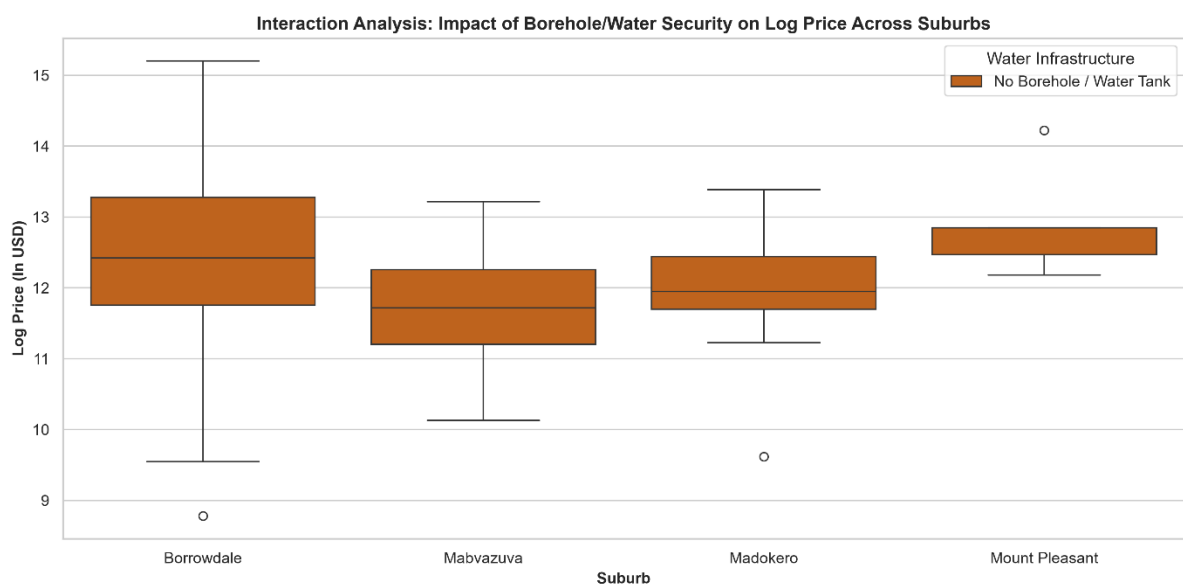
Diagnostic Breakdown:

- NaN VIFs on Text Flags:** In this run, the regex parser generated single-value vectors (all zeros) for the text flags across the dataset sample, resulting in zero variance ($\text{Var}(X) = 0$) and NaN values during covariance matrix inversion.
- 0.00 VIF on img_vis_painted:** Near-zero or zero variance in visual features indicates static predicted probability values across images.
- img_vis_paved VIF (4.00):** Within the safe operational threshold ($\text{VIF} < 5.0$).

Technical Resolution Prior to Modeling:

- **Constant Column Removal:** Drop all text features returning zero variance (txt_*) and visual features returning constant probabilities before passing features into OLS, Ridge, or Lasso models to prevent linear algebra singular matrix errors.
- **Regex Refinement:** Re-verify description text parsing rules to ensure text flags correctly extract presence/absence binaries (0 or 1) with non-zero variance.

4. Multimodal Condition Visualization Across Suburbs



Analysis of the Diagnostic Plot:

1. **Price Dispersion by Suburb:** Borrowdale exhibits the largest absolute price variance, with log-prices ranging from 8.8 ($\approx$ \$6,600, potential land/unbuilt plots) up to > 15.0 ($\approx$ \$3.2M luxury mansions).
2. **High-Density Tightness:** Planned suburban developments (Madokero and Mabvazuva) show narrow interquartile ranges (IQR) concentrated between log-prices 11.0 and 12.5 (\$60,000 – \$270,000).
3. **Outlier Identification:** Single lower-tail outliers are visible in Madokero (log-price ≈ 9.6 , representing entry-level shell structures) and an upper-tail outlier in Mount Pleasant (log-price ≈ 14.2 , representing prime commercial conversions).